

Retrieval-augmented generation: a hybrid approach to assessing retrieved documents similarity, LLM confidence, and system stability

Alexander Plesovskikh

aeplesovskikh@edu.hse.ru

Higher School of Economics

Short Report

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Retrieval-augmented generation: a hybrid approach to assessing retrieved documents similarity, LLM confidence, and system stability

Alexander Plesovskikh^{1*}

*Correspondence: aeplesovskikh@edu.hse.ru

¹Moscow Institute of Electronics and Mathematics, Higher School of Economics, 34, Tallinskaya St., Moscow, 123592, Russia

Abstract

The retrieval-augmented generation approach relies on retrieving relevant documents to enhance large language model prompts and improve model outputs. However, existing metrics like cosine similarity, precision@k, and recall@k, to name just a few, fail to account for the confidence and stability of retrieval and generation. We propose a novel approach, retrieval confidence score, and its extension, asymptotic retrieval confidence score, which combines semantic similarity, large language model confidence, and stability across multiple generations. Asymptotic retrieval confidence score potentially provides a robust approach for evaluating retrieval-augmented generation systems, possibly suggesting a better solution for combining results across retrieval, generation, and evaluation stages.

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List of abbreviations:

LLM – large language model

RAG – retrieval-augmented generation

Traditional retrieval-augmented generation metrics like recall@k or precision@k ignore LLM uncertainty in its responses and instability [1]. Poor retrieval can propagate errors in downstream tasks [2], e.g., question-answering tasks. In retrieval-augmented generation systems, the retrieval stage involves fetching relevant documents or information from an external knowledge source based on a user’s query [3]. The generation stage then uses this retrieved information, along with the query, to produce a coherent and accurate response, typically leveraging a large language model. In the survey paper [4], authors examined a variety of separate metrics to evaluate these two stages, such as precision and recall for retrieval and ROUGE or BLEU scores for generation (Table 1).

Table 1 Retrieval and generation metrics overview

Stage	Metric	Definition
Retrieval	Accuracy	The ratio of correct predictions (including both true positives and true negatives) to the total number of cases evaluated.
	Precision	The percentage of retrieved items that are actually relevant.
	Recall at k (Recall@k)	The percentage of truly relevant instances captured within the first k retrieved documents.
	Mean Reciprocal Rank (MRR)	The average of the inverse ranks of the first relevant result across multiple queries.
	Mean Average Precision (MAP)	The mean of the average precision scores for each query.

Generation	Recall-Oriented Understudy for Gisting Evaluation (ROUGE)	A family of automatic evaluation measures that quantify how well system summaries match human references.
	Bilingual Evaluation Understudy (BLEU)	An automatic metric that measures the similarity between machine-translated text and human reference translations.
	BERTScore	BERTScore evaluates text quality by comparing contextual embeddings of generated and reference texts using pre-trained language models like BERT.
	‘LLM as a Judge’	A more recent approach where large language models assess the quality of generated text (e.g., summaries, answers, translations) by scoring criteria like coherence, relevance, fluency, and factual consistency.

Information about the metrics in Table 1 is adapted from the study [4].

The goal of this study is to introduce a compound metric that assesses the entire RAG pipeline. Such a unified metric would capture the interplay between retrieval quality and generation effectiveness, addressing the limitations of isolated evaluations and providing a more comprehensive measure of overall system performance.

Retrieval confidence score (RCS)

Let an input query $q \in Q$ be given, where Q is the space of possible queries. The set of relevant documents $D_r = \{d_1, \dots, d_k\} \subset D$, where D is the corpus of documents.

Definition 1 (relevance measure)

The similarity function: $(q, d) \rightarrow [0, 1]$ specifies the semantic similarity [5] between a query (q) and a document (d), satisfying the following rules:

- $\text{similarity}(q, d) = 1 \Leftrightarrow q$ and d are semantically equivalent;
- $\text{similarity}(q, d) = 0.5 \Leftrightarrow q$ and d are semantically orthogonal;
- $\text{similarity}(q, d) = 0 \Leftrightarrow q$ and d are semantically dissimilar.

Definition 2 (confidence measure)

For fixed q and d , we perform n independent generations, where:

- $\mu(r|q, d) = \mathbb{E}[\text{conf}(r|q, d)]$ is a true confidence measure of an LLM response to a given pair of query (q) and document (d);
- $\sigma^2(r|q, d) = \text{Var}(\text{conf}(r|q, d))$ is a confidence variance.

RCS measures the expected LLM confidence in its response (r) to a user query (q), weighted by similarity between a query (q) and document (d) (1):

$$RCS(q, d, r) = \text{conf}(r | q, d) \text{similarity}(q, d)$$

where:

- $\text{similarity}(q, d)$ is a similarity function, usually it is a cosine similarity between the document retrieved and the query, but normalized from $[-1, 1]$ to $[0, 1]$: $\frac{1}{2}(\text{cosine similarity}(q, d) + 1)$, note that in general it is not limited solely to cosine similarity (e.g., Euclidean distance (L2), dot product similarity, etc.).

- $conf(r | q, d)$ is a normalized confidence score from the LLM for a response to a given document and query, note that in general, it is not limited to the LLM score (e.g., human evaluation, etc.)

Meanwhile, the asymptotic retrieval confidence score (ARCS) penalizes responses, causing instability in LLM confidence by incorporating variance in confidence over multiple generations (2):

$$ARCS(q, d, r) = conf(r | q, d) \cdot similarity(q, d) (1 - 2\sigma_{conf}(r | q, d))$$

where:

- $\sigma_{conf}(r | q, d)$ is a standard deviation of LLM confidence scores across multiple generations.

When an LLM is confident in its responses, i.e., its variance approaches 0, then $ARCS(q, d, r)_{|\sigma^2 \rightarrow 0} \rightarrow RCS(q, d, r)$. Furthermore, when LLM responses are appropriate, receiving high scores of 1 and documents receive high similarity scores given a user query, then $RCS(q, d, r) \rightarrow 1$, otherwise $RCS(q, d, r) \rightarrow 0$ if either retrieved documents are not similar or LLM is not confident in generated responses.

The proof-of-work in the study is demonstrated through a generation process involving 1000 queries across various-sized vector databases, Sentence Transformers, and large language models. Figure 1 (left) illustrates the scenario where different-sized Sentence Transformers or datasets are used, as the size of the dataset increases, the similarity between query embeddings and retrieved results increases; whilst Figure 1 (right) shows the scenario where LLM confidence increases as the dataset grows and model's ability to generate a response based on query and retrieved documents.

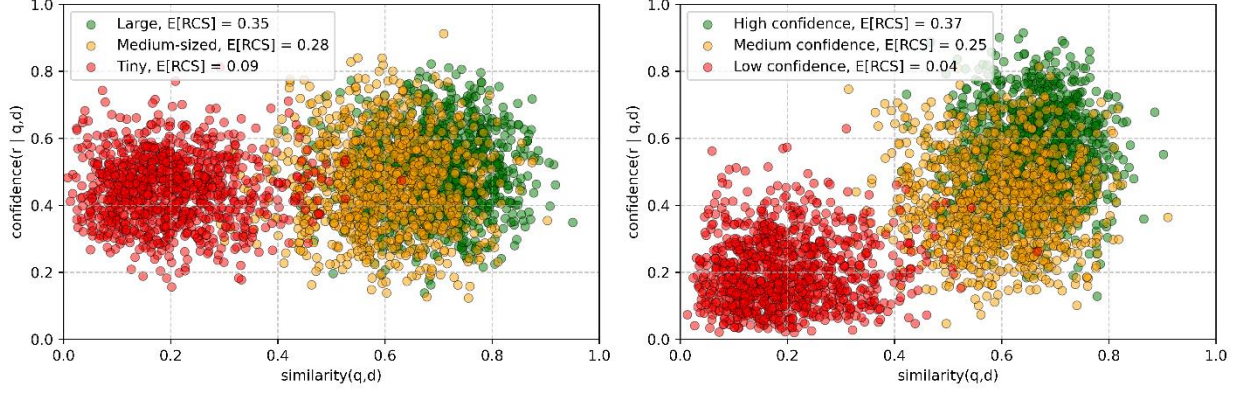


Fig. 1 RCS scatter plot: (Left) Similarity increases with larger datasets; (Right) LLM confidence increases

The ARCS metric combines three factors that are the following: semantic similarity, which measures how well a document matches a user query, LLM confidence, which assesses the model's certainty in the response correctness to a user query, and stability, which penalizes high variance in confidence across multiple generations. The proposed approach works with any embedding model that provides a similarity score in the normalized $[0, 1]$ range, and it can be applied to different LLM models. The confidence parameter is either self-assessed by LLM after generating a response or by a group of different LLMs. This metric filters out not only unrelated documents, as it does precision@k or recall@k and a bunch of other measures, but also confidence and stability of scores. The proposed metric has an interpretable score range from 0 to 1, where 1 means a perfect match, high confidence in responses and stable LLM response across multiple generations, and 0 when either retrieved documents are not similar or LLM is not confident in generated responses or LLM is hugely uncertain by the theoretically maximum amount of deviation.

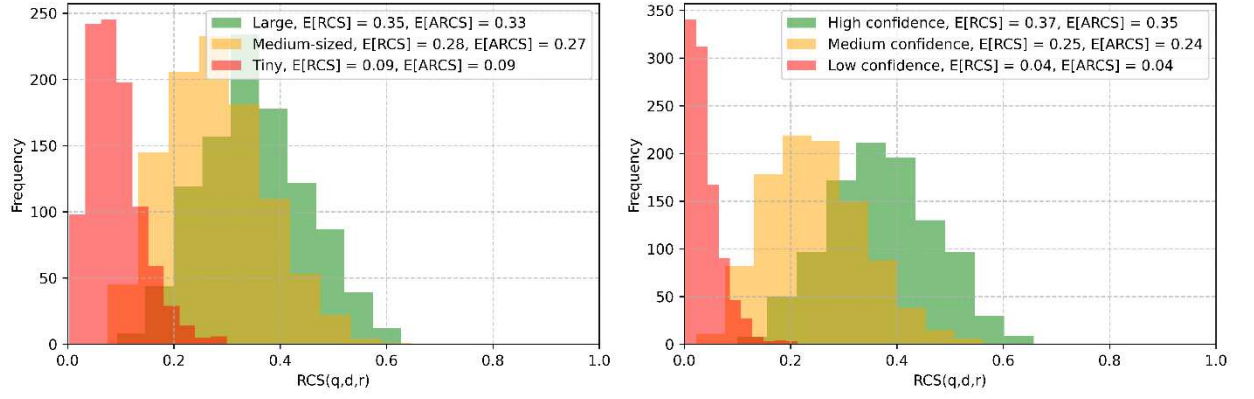


Fig. 2 ARCS accounts for high variance in confidence across multiple generations

Meanwhile, we have to address the cons of such a metric. Firstly, it requires multiple LLM generations to compute confidence variance, which could be computationally expensive, but on the other hand, it gives an opportunity to leverage a number of generations, choosing either high performance or high confidence in metrics [5]. The RSC may penalize some documents that are inherently uncertain, e.g., in subjective topics. Moreover, the effectiveness of LLM-based scores can vary, and there’s no universally applicable grading scale and prompting text, complicating the standardization of ‘LLM as a Judge’ [4]. Nevertheless, the use of ‘LLM as a Judge’ is not restricted to this specific case; it can also incorporate human experts. Furthermore, the robustness of confidence scoring can be enhanced by increasing the number of evaluations (n), conducted by either human assessors or several LLMs, to improve reliability.

Author contributions

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Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare that they have no competing interests.

References

1. Abbasi YY, Kuzborskij I, György A, Szepesvari C. To believe or not to believe your llm: Iterative prompting for estimating epistemic uncertainty. *Advances in Neural Information Processing Systems*. 2024;37:58077-117.
2. Cuconasu F, Trappolini G, Siciliano F, Filice S, Campagnano C, Maarek Y, Tonellotto N, Silvestri F. The power of noise: Redefining retrieval for rag systems. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*. 2024;719-29.

3. Li Z, Wang Z, Wang W, Hung K, Xie H, Wang FL. Retrieval-Augmented Generation for Educational Application: A Systematic Survey. *Computers and Education: Artificial Intelligence*. 2025;100417.
4. Yu H, Gan A, Zhang K, Tong S, Liu Q, Liu Z. Evaluation of retrieval-augmented generation: A survey. In *CCF Conference on Big Data 2024 Aug 8* (pp. 102-120). Singapore: Springer Nature Singapore.
5. Han J, Kamber M, Pei J. 2 - Getting to Know Your Data. In: Han J, Kamber M, Pei J, editors. *The Morgan Kaufmann Series in Data Management Systems, Data Mining: Concepts and Techniques (Third Edition)*. Morgan Kaufmann; 2012. P. 39-82.
6. Yang X, Sun K, Xin H, Sun Y, Bhalla N, Chen X, Choudhary S, Gui R, Jiang Z, Jiang Z, Kong L. Crag-comprehensive rag benchmark. *Advances in Neural Information Processing Systems*. 2024;37:10470-90.